

Cybersecurity Concepts Lab for Teachers

Trust your hackles. Pause · Check · Tell.

Trust your hackles. Pause · Check · Tell.

FIELD COPY

1



Lock It

PASSWORDS AND ACCOUNTS

2



Spot It

SCAMS AND FAKES

3



Share Smart

PRIVACY

4



Safe People

WHO TO TRUST

5



Kind and Well

KINDNESS AND BALANCE

6



Fix It

TELL, REPAIR, RECOVER

PLATE I. The six strands, hand-inked, in the order the lessons take them.

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OCTOBER 2026

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Demo 3: Anatomy of a Phish (Annotate the Fake)

Trust your hackles. Pause · Check · Tell

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Status: DRAFT for teacher, counselor and legal review. Items marked [VERIFY] need checking against a primary source before publication.

AT A GLANCE	
Big idea	A phish is a message built to make you act before you think. Judge the ask and the path , not how professional it looks.
Grade bands	K-2 (the "fishing lure" game), 3-5 (full), 6-8 (full + taxonomy), 9-12 (full + modern variants and reporting)
CSTA-style concepts	Networks and the Internet (cybersecurity), Impacts of Computing (safety, privacy) (codes: see Concepts Lab Index, Part 4 [VERIFIED 2026-10-05])
Time	15-minute and 45-minute versions
Cost and tech	Printed fake messages (provided below), highlighters, sticky notes. No devices.
Teachers do it first	Yes. Part A is about 20 minutes.

MAKE IT STICK

Try This Now (60 seconds, no prep): Read this fictional text aloud: "Parcel Pal: your package could not be delivered. Pay a 1.99 redelivery fee: `parcelpal-fee.test/pay`". In 20 seconds, find the pressure, the payload (the link) and the ask. Notice you did not need to say "it looks fake".

01 The big idea

"Phishing" is a trick where someone pretends to be someone you trust, such as a bank, a school, a game, a boss or a friend, to get you to click, share a secret, send money or install something. The word is a play on "fishing": a lure (the message) is dangled, and the person who bites gets hooked.

The most important teaching shift in this demo: **stop asking "does it look fake?" and start asking "what is it asking me to do, and how did it reach me?"** Modern messages can be polished, grammatically perfect and carry real logos, especially now that AI can write fluent text. Spelling mistakes are not a reliable test. The ask is.

02 What a teacher should know (plain language)

2.1 WHY PHISHING WORKS (IT IS NOT ABOUT BEING GULLIBLE)

PHISHERS USE ORDINARY HUMAN shortcuts. We respond fast to **urgency** (your account closes today), **authority** (a principal, a police officer, a boss), **fear** (you are in trouble), **greed or delight** (you won), **helpfulness** (a friend needs a favor) and **curiosity** (a photo of you?). Any of us can be hit on a tired Tuesday. The defense is a **habit** (pause, check, tell), not a personality trait. If someone is tricked, it is never their fault; the quicker they tell, the more can be fixed.

2.2 THE RED-FLAG TAXONOMY: "THE SIX P'S"

Use these six families on a wall poster. Every specimen below contains one or more.

P	WHAT IT LOOKS LIKE	EXAMPLE PHRASE	WHY IT IS A RED FLAG
Pressure	Urgency, countdowns, threats	"Act in 24 hours or your account is closed."	Pressure shuts down thinking. Real organizations rarely demand instant action through a link.
Prize	Too good to be true, free stuff	"You were selected for a free game currency drop."	You can't win a contest you never entered.
Pretender	Pretends to be someone you know or an authority	"This is your principal. I need a favor."	The sender name is easy to fake; trust must be checked by another route.
Path	Strange route: new number, unusual channel, link text vs real destination mismatch, secrecy	"Don't tell anyone, just click."	Real organizations use their official route; secrecy is a hallmark of manipulation.
Payload	A link, an attachment or a QR code you did not expect	"Open the attached invoice."	The clickable object is where the harm is delivered.
Payment or Passcode request	Asks for money (especially gift cards, crypto, wire), passwords, or one-time codes	"Read me the code you just got."	Legitimate organizations do not ask you to read out a one-time code or pay by gift card.

A seventh quick check belongs to everyone: **Unexpected?** If you were not waiting for this, treat it with extra care.

2.3 WHAT PHISHING LOOKS LIKE IN 2026 (SO TEACHERS ARE NOT STUCK IN 2010)

VARIANT	WHERE	NOTE
Email phishing	Inbox	Still the most common. Mass versions and carefully targeted versions ("spear phishing").
Smishing	Text/SMS and messaging apps	Package-fee, bank-alert and "wrong number" texts.
Vishing	Voice calls (including AI-cloned voices, see Demo 12)	Caller ID can be faked.
Quishing	QR codes	A QR code is just a link in a square; you cannot see where it goes by looking.
Account-takeover messages	A real friend's hacked account sends you a link	The sender is real; the account has been taken over. Message <i>content</i> is the clue.
Business email compromise	Fake boss or vendor asks for money or gift cards	In the FBI IC3 2025 report, business email compromise was the second-largest loss category (\$3.05 billion), after investment fraud (\$8.65 billion) [VERIFIED 2026-10-05: ic3.gov 2025 Internet Crime Report].
Consent phishing	A fake "grant this app access" screen	The app asks for broad permissions instead of a password.

Do not teach students to memorize every variant. Teach the six P's and the habit.

2.4 SAFE CHECKING HABITS (DEFENSE)

- 1. Do not use the link or number in the message.** Go to the organization by a route you already trust: your saved bookmark, the official app, the number on the back of your card, or a typed address.
- 2. Hover (on a computer) or press-and-hold (on a phone) to preview a link's destination** instead of tapping. The visible text can differ from the real address. Note: this is a clue, not a guarantee.
- 3. Check the sender through a second channel**, for example call your friend or ask the teacher in person.
- 4. Never share a one-time code, password or recovery code.** Whoever asks is not a legitimate helper.
- 5. If you clicked or shared something:** tell a trusted adult right away; change the password; if it was a code or payment, contact the bank or provider. Speed helps, shame does not.
- 6. Report it.** Use the "report phishing" button in your email app; forward text scams to 7726 (SPAM) in the US [VERIFIED 2026-10-05: consumer.ftc.gov]; in the US you can report to the FTC (ReportFraud.ftc.gov) [VERIFIED 2026-10-05: consumer.ftc.gov]; the

Anti-Phishing Working Group accepts forwarded phishing emails [APWG's Q4 2024 report asks for suspicious texts at apwg.org/sms (VERIFIED 2026-10-05); the email address was not checked, so look it up on apwg.org]. At school, report to your teacher or IT per school policy.

2.5 COMMON MISCONCEPTIONS

MYTH	BETTER UNDERSTANDING
"Phishing emails are full of typos."	Often not. AI-written lures can be flawless.
"If it has a padlock, it's safe."	The padlock shows the connection is encrypted, not that the site is honest (Demo 5).
"I'd never fall for it."	Everyone can. Experts, too, when tired or rushed. The habit is the defense.
"I should delete it and say nothing."	Report it. Others may have received the same one.
"All unexpected messages are scams."	No. That is why we check through a trusted route rather than assume.

03 Materials (zero to low cost)

- The **Specimen Cards** (Section 5, printed from the student handout pages; the answer notes are on the separate teacher page). All brands and people are fictional; web addresses use the reserved ".test" ending and invented brands ("Brightway Bank", "Nimbus Games", "Parcel Pal", "Greenfield School").
- Six highlighter colors (one per P) or sticky dots, a Six P's poster, index cards for the "Check Route" role-play, the handout.

04 Part A: Teachers do it first (about 20 minutes)

1. **Cold read (4 min).** Without the answers, mark which of the eight specimens you think are scams and which are real. Record your gut feeling.
2. **Annotate (8 min).** Highlight the six P's; circle the *ask* in red.
3. **Check the check (4 min).** For each, write the safe way you would verify (for example, "open my bank app directly").
4. **Compare (4 min).** Use the teacher answer page. Specimen G (real and a little annoying) tests over-suspicion; Specimen F (a friend's hacked account) tests misplaced trust in familiar names.

05 Specimen Cards (fictional; for annotation only)

THE EIGHT SPECIMENS (A to H) are printed on the student handout and are answer-free, so you can copy those pages as they are. Print one set per group. **Do not ask students to write their own deceptive messages** (see Safety). Answers and safe checks are on the teacher answer page.

06 Part B: Teaching it

15-MINUTE VERSION

MINUTES	WHAT HAPPENS
0-2	Hook (teacher script). Show the Six P's poster.
2-4	Model: annotate Specimen A together on the board.
4-10	Stations or table groups annotate Specimens B-F with six colors. Include G as the "real" one.
10-13	Reveal: groups share one P they found and one safe check.
13-15	Pause · Check · Tell exit slip (say it aloud: "I'll check by ____ and tell ____"): "I got a weird message; the first thing I do is ____."

45-MINUTE VERSION

MINUTES	WHAT HAPPENS
0-5	Cold-read warm-up: students vote real/fake by gut on three specimens, then reveal that their gut was wrong on at least one.
5-15	Teach the Six P's and psychology (why urgency works). Annotate A together.
15-30	"Phish Lab": groups annotate B-H, rank them by danger, and write the safe check for each on sticky notes.
30-38	"Check Route" role-play: pairs practice calling a trusted number (a card with fictional number) and saying a script. One partner is "the helpful adult".
38-43	Gallery walk: groups compare rankings, discuss Specimen G (the real one) to practice calm rather than paranoia.
43-45	Exit ticket and handout.

07 Teacher script

Hook (2 minutes). "Raise your hand if you've ever gotten a message that gave you a funny feeling. That feeling has a name in our kit: your hackles rising. Today we're going to take apart fake messages, like a biologist dissecting a specimen. We never make real tricks and we never aim anything at real people. We just learn to see the parts. Here's the amazing secret: the best clue isn't spelling. It's what the message wants you to do."

Model (annotate A). "Look at this: 'within 24 hours.' That's pressure. Mark it in orange. 'Click here.' Payload. 'Reply with your password.' A bank will

never ask for that. Passcode request. Notice we didn't have to say 'it looks fake.' We looked at what it wants."

Check Route. "When a message gets your hackles up, you pause. You check by a different road: the bank's own app, a number you already have, a grown-up. You tell. If you clicked, you tell fast. A fast tell is a brave tell."

Close. "If a trick ever works on you, it's never your fault. These tricks are built by professionals to work on everyone. The only losing move is hiding it."

08 Kid-friendly analogies by band

BAND	ANALOGY AND ACTIVITY
K-2	The lure game. Pretend fishing: a toy fish with a shiny lure (a "free toy! click here!"). The class decides "bite or pause?" Practice the signal: hand up the back of the neck (Hackles Up), then Pause Palm, then Tell Wave. Use drawn cartoon messages, not real ones.
3-5	The disguise party. The message is wearing a costume. Look under the costume at what it asks. Annotate with colored dots. Practice a "Check Route": ask a trusted adult, use the app, not the link.
6-8	The Six P's taxonomy. Annotate, rank, discuss why professionals still fall for it, practice reporting and "two-channel verification". Debate: Why is secrecy a red flag?
9-12	Attack surface and incentives. Variants (smishing, quishing, vishing, consent phishing), how stolen data makes lures believable, MFA and passkeys as phishing defense (Demo 6), organizational reporting and incident response (Demo 13).

09 Discussion questions

1. Which red flag is easiest for scammers to remove? Which is hardest? (Hint: spelling vs. urgency and the ask.)
2. Why do scammers like gift cards and secrecy?
3. Why might a message from a real friend still be unsafe?
4. (9-12) How do AI writing tools change the red flags we can rely on?

10 Level-up challenge

- **Safer by design.** Rewrite Specimen A into a *legitimate* security alert that has no link, says "open the app yourself" and avoids pressure.
- **Class "Check Route card."** A wallet card with trusted numbers, official app names and report addresses (generic info only, no personal data).
- **Case study (9-12).** Map a publicly documented phishing case from a government source to the six P's [VERIFY a suitable CISA or FBI public advisory].

11 Safety and ethics notes (defense only)

- **No attack how-to.** Do not teach students to compose realistic phishing messages, clone sites or send test messages to real people. The level-up "Safer by design" is the opposite: build the safe version.
- Use only the fictional specimens or ones you have de-identified. Do not show real malicious links; never ask students to click any link in class.
- Do not run "phishing simulations" on students, parents or staff without explicit policy, notice and consent.
- Do not use real organization names, real principals or real students in specimens.
- Reduce shame: the lesson is about a pattern, not about being tricked. If a student discloses that they have been tricked, threatened or blackmailed (for example "someone said they will share my photos"), follow the school's reporting procedure; see the box in the Index and your Teacher Guide's disclosure section. Do not ask the student to show you images.

12 Assessment idea

- **Specimen exit ticket.** Show a new fictional specimen. Students list the P's present and one safe check. Rubric: 3 = identifies 2+ P's and a check that avoids the message's link; 2 = identifies one flag; 1 = says "it looks fake" with no reason.
- **Sorting.** Four real and four fake messages; students sort and justify. At least one real message must be included to measure overcorrection (paranoia).

13 Sources and [VERIFY] list

- FBI Internet Crime Complaint Center (IC3) 2025 annual report: phishing/spoofing was the most-reported crime type (191,561 complaints) and BEC the second-largest loss category [VERIFIED 2026-10-05: ic3.gov].
- 7726 (SPAM) text reporting in the US [VERIFIED 2026-10-05: consumer.ftc.gov].
- FTC ReportFraud and consumer.ftc.gov scam guides [VERIFIED 2026-10-05].
- Anti-Phishing Working Group trend reports; the exact share of phishing sites that use HTTPS was not verified and no percentage is claimed (see Demo 5).
- CISA "Avoiding Social Engineering and Phishing Attacks" guidance [page exists: VERIFIED 2026-10-05: cisa.gov].



- Research on the limitations of "spot the typo" training and simulated-phishing click-rate metrics [VERIFY current studies before citing].

SECTION

1

Student Handout: Anatomy of a Phish

Name: _____

The big idea

A **PHISH TRIES TO** make you act before you think. Look at **what it asks** and **how it reached you**, not just whether it looks fake.

The Six P's

P	WHAT IT MEANS
Pressure	Hurry! Act now! Or else!
Prize	Too good to be true, free stuff
Pretender	Pretends to be someone you trust
Path	Strange route or secrecy; link does not match
Payload	A link, attachment or QR code you did not expect
Payment or Passcode	Asks for money, gift cards, passwords or one-time codes

Warm-up

"Nimbus Games: Congratulations! You're chosen for a FREE skin pack. Log in at nimbus-skins.test in the next 10 minutes. Don't tell anyone or the offer ends!"

Circle the **ask**. P's I found: _____ My safe check:

Specimen Cards (all made up; never click anything in real life)

FOR EACH ONE: CIRCLE the **ask**, write the P's you find, and write your safe check in the table at the end.

A. Email. From: Brightway Bank Security <alerts@brightway-secure.test>. Subject: Final warning: your account will be suspended. "Dear Customer, we noticed unusual activity. You must verify your identity within 24 hours or your account will be closed permanently. Click here: <http://brightway-login.test/verify>. Please reply with your password and the code we texted you."

B. Text message. "Parcel Pal: your package could not be delivered. Pay a 1.99 redelivery fee: parcelpal-fee.test/pay"

C. Game chat. "Hey! You've been picked for a free Nimbus Games Mega Pack! To claim, log in at nimbus-megapack.test. Be quick, only 100 left. Don't tell your friends yet, they'll take them!"

D. "The principal." From: "Principal Dana Reyes" <dreyes.greenfield@mailbox.test>. "I'm in a meeting and can't talk. I need a quick favor, urgent and confidential. Please buy four 50-dollar gift cards and send me the codes. I'll pay you back."

E. QR code poster. A poster taped to a school bulletin board: "Scan for free pizza for your class!" with a QR code and no organization named.

F. A friend's account. From a friend's real chat account: "omg is this you in this video??? greenfield-clip.test/watch"

G. Library notice. "Greenfield School Library: Your book 'Ocean Mysteries' is due on Friday. Log in to our library catalog as you usually do to renew."

H. Pop-up. A window flashes: "WARNING: Your computer is infected! Call this number now: 555-0100 or all your photos will be deleted!"

SPECIMEN	P'S I FOUND	REAL OR SCAM?	MY SAFE CHECK
A			
B			
C			
D			
E			
F			
G			

SPECIMEN	P'S I FOUND	REAL OR SCAM?	MY SAFE CHECK
H			

My Check Route

IF A MESSAGE GIVES me a funny feeling, I will:

1. **Pause:** no clicking, no replying.
2. **Check:** use the real app or a number I already know; ask a trusted adult.
3. **Tell:** a trusted adult, every time.

My trusted adults: _____

Remember

- Spelling mistakes are not a reliable clue. The ask is.
- Never share a one-time code or password with anyone who contacts you.
- A false alarm is fine and welcome.
- If you clicked or shared something: **it is not your fault.** Tell a trusted adult right away.

End of student handout.

TEACHER COPY ONLY:

Answers for Demo 3 (do not distribute to students)

FIELD NOTE

No. 1

WARNING: Teacher page. Remove before copying. The student handout ends on the page before this one. Print the handout pages only, or detach this page first. Do not post it, project it or send it home.

Warm-up (Nimbus skin pack): Prize, Pressure (10 minutes), Pretender (pretends to be the game), Path (secrecy), Payload (the login link). Safe check: only claim rewards inside the real game, and ask an adult.

Specimen answers

A, Brightway Bank email (scam). Pressure (24 hours, "final warning"), Pretender (looks like a bank), Path (sender and link addresses do not match the bank's known site), Payload (link), Passcode request (password and the texted code). Safe check: ignore the link; open the bank's app or call the number on the back of your card. Report it.

B, Parcel Pal text (scam). Pressure, Payload (link), Payment. Also Unexpected: if you are not waiting for a package, it is a flag; even if you are, use the carrier's app. Forward to 7726 [VERIFIED 2026-10-05: consumer.ftc.gov] and delete.

C, Game chat (scam). Prize, Pressure (only 100 left), Path (secrecy), Payload, Pretender (pretends to be the game). Safe check: only claim rewards inside the real game app; tell an adult; report the user in the game.

D, "The principal" (scam). Pretender (authority), Pressure, Path (personal address, secrecy, "can't talk"), Payment (gift cards are a classic scam payment because they are hard to trace). Safe check: walk to the office or call a number you already know; never buy gift cards on request. Fictional name; never use a real principal's name.

E, QR poster (scam risk). Prize, Path (unknown source), Payload (QR code). A QR code is just a link in a square. Safe check: ask a teacher; only scan codes from sources you can identify.

2

F, Friend's account (scam: likely a taken-over account). Payload, Pressure (curiosity), Pretender (the friend's account may be taken over), Path (odd phrasing and a link). Safe check: message your friend by another route, "Did you send that?" Do not click.

G, Library notice (real, the control). No ask beyond what you already do; no link; no pressure or prize; expected. Real and boring. If unsure, still use the usual route.

H, Pop-up (scam). Pressure and fear, Path (a pop-up, a number supplied by the pop-up), Payment risk (a "support fee"). Safe check: close the browser, do not call, tell an adult and let the adult or IT handle any scan.

Scoring tip: accept any reasonable P list; the goal is naming the ask and choosing a safe check that avoids the message's own link or number. Check that students did not label G a scam by reflex.